

1.10 Early Finishers: Sequences and Compound Interest

For students who finish the quiz early. A calculator is allowed. Show your working: write the formula you use, then substitute.

1. In an arithmetic sequence, $u_3 + u_8 = 48$ and $u_{12} = 5u_2$.
 - (a) Find the first term u_1 and the common difference d .

- (b) Find the number of terms of this sequence that are less than 500.

A different arithmetic sequence has first term 3 and common difference d , where $d \neq 0$. Its 1st, 2nd and 4th terms are the first three terms of a geometric sequence.

- (c) Find the value of d and the common ratio of the geometric sequence.

Nickname:

2. Sofia invests \$5000 in an account that pays a nominal annual interest rate of 4.8%, compounded monthly.

- (a) Find the least number of whole years after which Sofia's investment is worth more than \$10 000.

At the same time, Luca invests \$6000 in an account that pays a nominal annual interest rate of $r\%$, compounded quarterly. After 10 years Luca's investment is worth \$9000.

- (b) Find the value of r .

- (c) Find the first whole year at the end of which Sofia's investment is worth more than Luca's.